ENTERPRISE SUPPORT OPERATIONS | CONTROLLED KNOWLEDGE BASE

Enterprise Complaint Routing Knowledge Base

Operational resolution procedures, diagnostics, routing guidance and approved resolver playbooks

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Knowledge Base ID	KB-ESOPS-001
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PURPOSE
This knowledge base is the operational reference for resolving common complaint patterns in the enterprise complaint-routing platform. It is structured for human resolver use and later retrieval by the AI resolution assistant.

GOVERNANCE
Articles are approved operational guidance. They do not authorize irreversible actions, credential disclosure, financial adjustments or security exceptions without appropriate policy and approval.

Prepared for: Customer Support, Service Desk, Resolver Teams, Operations Managers, Platform Administrators and AI-assisted Support Operations

1. Knowledge Base Operating Model

This knowledge base is designed as a controlled, article-based operational library. Each article has a stable identifier, a defined audience, symptoms, likely causes, safe procedures, validation criteria and escalation conditions. The same structure supports human search, semantic retrieval and future GraphRAG or retrieval-augmented generation workflows.

1.1 How to use an article

- Identify the ticket symptom and confirm that the article applies before executing steps.
- Use the current ticket data, SLA state, category, subcategory, customer impact and existing audit trail as the primary context.
- Execute only steps authorized for the resolver role. Do not invent system facts or claim that a remediation succeeded without verification.
- Record material actions in the ticket timeline or internal comment stream where available.
- Escalate when the article says to escalate, when evidence contradicts the article, or when the ticket crosses an approval boundary.

1.2 Article taxonomy

Domain	Typical article IDs	Primary resolver use
Identity & Access	KB-ACC-001 to 003	Login, password, account lockout, identity compromise
Security & Privacy	KB-SEC-001 to 003	Suspicious login, data exposure, unauthorized access
Billing & Subscription	KB-FIN-001 to 003	Duplicate charges, payment, subscription activation/refunds
Technical & Availability	KB-TECH-001 to 004	Service outage, connectivity, integration/API issues
Platform Operations	KB-OPS-001 to 004	Routing, SLA, assignment, notification and workflow problems
AI-assisted Resolution	KB-AI-001 to 002	AI recommendation use and similar-case analysis

1.3 Retrieval fields

Recommended metadata for indexing: article_id, title, domain, product_area, symptom, category, subcategory, priority, resolver_role, keywords, prerequisites, procedure, validation, escalation, related_article_ids, effective_date, owner, version.

RETRIEVAL PRINCIPLE
The best retrieval result is not necessarily the most similar sentence. Prefer an article with the correct category, subcategory, resolver role and operational condition, then use text similarity to rank within that filtered set.

2. Priority, SLA and Routing Decision Reference

Use this page as the fast operational matrix before opening a detailed article. Final priority remains subject to the platform priority engine and authorized operational override.

Priority	Typical business condition	Default response posture	Resolver guidance
Critical	Security compromise, broad outage, severe business stoppage, major financial exposure	Immediate triage; shortest SLA; escalation readiness	Use experienced resolver, prefer strongest skill fit and appropriate on-call coverage
High	Material business impact, customer-blocking failure, significant payment/subscription problem	Prompt assignment; monitor SLA closely	Prefer skilled resolver with available capacity and healthy SLA exposure
Medium	Contained operational issue, common account or billing issue	Standard queue handling	Use balanced workload and skill fit
Low	Information, appreciation, minor issue, non-blocking request	Normal queue; avoid overloading critical specialists	Use general technical/support capacity where appropriate

2.1 Resolver assignment gates

- Correct operational team is a hard boundary for automatic routing.
- Resolver must be active and configured for automatic assignment.
- Normal automatic routing requires workforce eligibility, schedule alignment and capacity availability.
- Skill fit influences ranking; it does not replace a hard eligibility rule.
- Priority load and SLA exposure reduce the score of heavily burdened resolvers.
- On-call routing is an emergency fallback for defined priority conditions when normal workforce candidates are unavailable.
- Manual admin assignment is an operational override and must be auditable.

2.2 Ticket lifecycle

State	Meaning	Required ownership
OPEN	Ticket exists; may be unassigned or assigned and waiting to be started	No resolver required for an unassigned queue ticket
IN_PROGRESS	Resolver has actively started work	Assigned resolver required
PENDING_CUSTOMER	Resolver is waiting for customer information/action	Assigned resolver required
RESOLVED	Resolver has completed the technical/service resolution	Assigned resolver and resolution evidence expected
CLOSED	Ticket is administratively closed after resolution/confirmation	Historical record

3. KB-ACC-001 - Account locked / unable to sign in

Use when a customer cannot access the application because the account appears locked or login attempts are rejected.

Field	Value
Article ID	KB-ACC-001
Audience	Customer Support, Technical Support, Resolver
Severity / Priority	Medium to High
Applies when	Customer reports locked account or repeated login rejection.
Related articles	KB-SEC-001, KB-TECH-001

Resolution context

- Symptom: Customer reports locked account or repeated login rejection.
- Symptom: No evidence yet of malicious account takeover.
- Likely cause: Account lockout policy triggered.
- Likely cause: Incorrect credentials or stale browser/session state.
- Likely cause: Identity synchronization or authentication service issue.

Prerequisites and guardrails

- Confirm customer identity using approved verification controls.
- Do not request or store passwords, MFA secrets or recovery codes.
- Review ticket history for recent password or security changes.

Resolution procedure

1. 1. Confirm exact error text and approximate start time.
2. 2. Check whether other users in the same tenant are affected.
3. 3. Verify whether the identity provider/authentication dependency is operational.
4. 4. If the issue is a local lockout, use the authorized account-unlock workflow.
5. 5. Ask the customer to retry through a clean session or approved client path.
6. 6. If the error persists across users or tenants, move to technical incident analysis.

Validation and closure criteria

- Customer can authenticate successfully.
- No active lockout remains.
- Ticket contains verification evidence and final action taken.

Escalation

- Escalate to Cyber Security when takeover indicators appear.
- Escalate to Site Reliability when multiple users/tenants are simultaneously affected.
- Escalate to platform owner when identity synchronization remains inconsistent after approved remediation.

OPERATOR NOTE
Never bypass identity verification merely to meet SLA.

4. KB-ACC-002 - Password reset email not received

Use when password reset was requested but the verification/reset email does not arrive.

Field	Value
Article ID	KB-ACC-002
Audience	Customer Support, Technical Support, Resolver
Severity / Priority	Medium
Applies when	Customer can access email but no reset message is received.
Related articles	KB-ACC-001, KB-OPS-002

Resolution context

- Symptom: Customer can access email but no reset message is received.
- Symptom: Reset request may succeed in the application but email is absent.
- Likely cause: Mailbox filtering or spam quarantine.
- Likely cause: Email delivery delay.
- Likely cause: Incorrect customer email or stale identity record.
- Likely cause: Notification provider outage.

Prerequisites and guardrails

- Confirm the destination email without exposing private account data.
- Check whether the customer has received any other system notifications.
- Do not disclose whether an account exists to an unverified party.

Resolution procedure

7. 1. Confirm request timestamp and channel.
8. 2. Ask customer to check spam/quarantine and corporate mail filters.
9. 3. Verify the application generated the reset request successfully.
10. 4. Retry only through the approved reset mechanism.
11. 5. If repeated attempts fail, inspect notification delivery telemetry or notification status.
12. 6. Escalate repeated delivery failures as a platform/notification incident.

Validation and closure criteria

- Customer receives a valid reset message.
- Customer can complete password reset.
- No duplicate or unsafe reset tokens remain active.

Escalation

- Escalate to notification/platform operations for repeated delivery failure.
- Escalate to security if the reset request appears unauthorized.

OPERATOR NOTE
Avoid telling the customer that an account does or does not exist based solely on email delivery behavior.

5. KB-SEC-001 - Suspicious login alert / suspected credential compromise

Use when a customer reports foreign login alerts, suspicious sessions or possible credential compromise.

Field	Value
Article ID	KB-SEC-001
Audience	Cyber Security, Fraud Investigation, Resolver
Severity / Priority	Critical
Applies when	Unexpected login from an unfamiliar location.
Related articles	KB-SEC-002, KB-FIN-001

Resolution context

- Symptom: Unexpected login from an unfamiliar location.
- Symptom: Customer believes credentials were compromised.
- Likely cause: Credential theft.
- Likely cause: Session/token compromise.
- Likely cause: Compromised endpoint or browser.

Prerequisites and guardrails

- Treat as a potential security incident.
- Use the approved identity verification process.
- Do not ask for passwords, MFA codes or private recovery secrets.

Resolution procedure

13. 1. Capture the alert time, location and observed session information available to the customer.
14. 2. Determine whether the customer still controls the registered communication channel.
15. 3. Restrict or invalidate sessions using the approved security workflow when authorized.
16. 4. Force credential reset through the approved mechanism.
17. 5. Review recent account activity for unauthorized changes or transactions.
18. 6. Route financial-loss indicators to Fraud Investigation; escalate material or multi-account events to security incident handling.

Validation and closure criteria

- Unauthorized sessions are contained.
- Credential reset is completed through approved controls.
- Any financial or data-impact findings are captured and routed.

Escalation

- Immediate escalation for active compromise, privileged access, data exposure or financial loss.
- Escalate to Fraud Investigation for unauthorized payments.

OPERATOR NOTE
Speed matters, but evidence handling and identity verification remain mandatory.

6. KB-SEC-002 - Customer sees another customer's personal information

Use when a customer reports cross-customer visibility or data exposure.

Field	Value
Article ID	KB-SEC-002
Audience	Cyber Security, Privacy, Operations Manager
Severity / Priority	Critical
Applies when	Customer can view another user/customer record.
Related articles	KB-SEC-001, KB-OPS-004

Resolution context

- Symptom: Customer can view another user/customer record.
- Symptom: Screenshot or evidence indicates unauthorized data access.
- Likely cause: Authorization policy defect.
- Likely cause: Tenant/row-level data isolation failure.
- Likely cause: Cache or session contamination.
- Likely cause: Configuration or deployment regression.

Prerequisites and guardrails

- Treat as a potential privacy/security incident.
- Preserve evidence and avoid broad testing with real customer data.
- Do not reproduce by accessing unrelated customer records unless specifically authorized.

Resolution procedure

19. 1. Capture the minimum necessary evidence from the reporter.
20. 2. Identify tenant, user context and approximate time.
21. 3. Stop further exposure when an approved containment action exists.
22. 4. Escalate immediately to Cyber Security/Privacy incident handling.
23. 5. Record affected data classes without copying unnecessary sensitive values.
24. 6. Coordinate remediation and customer communication through incident ownership.

Validation and closure criteria

- Exposure is contained.
- Incident owner confirms remediation.
- Required notifications and audit records are complete.

Escalation

- Immediate security/privacy escalation.
- Operations manager approval required for customer-impacting containment changes.

OPERATOR NOTE
This article is not a routine resolver troubleshooting procedure; follow incident-management controls when exposure is confirmed.

7. KB-FIN-001 - Duplicate charge / duplicate payment

Use when a customer reports being charged twice for the same purchase or subscription.

Field	Value
Article ID	KB-FIN-001
Audience	Finance Operations, Resolver
Severity / Priority	High
Applies when	Two or more transactions appear to represent the same intended purchase.
Related articles	KB-FIN-002, KB-SEC-001

Resolution context

- Symptom: Two or more transactions appear to represent the same intended purchase.
- Symptom: Customer requests reversal or refund.
- Likely cause: Duplicate submission.
- Likely cause: Payment retry after timeout.
- Likely cause: Gateway duplication.
- Likely cause: Delayed authorization/capture creating apparent duplicates.

Prerequisites and guardrails

- Use transaction identifiers, not card secrets.
- Do not promise a refund before transaction status is verified.
- Follow finance approval limits for refunds/reversals.

Resolution procedure

25. 1. Capture transaction date, amount, currency and non-sensitive reference identifiers.
26. 2. Determine whether both transactions are settled, pending or reversed.
27. 3. Check for duplicate order/subscription records.
28. 4. If one transaction is only an authorization, explain the expected release path.
29. 5. If both are settled duplicates, initiate the approved refund/reversal workflow.
30. 6. Record the finance action and reference number in the ticket.

Validation and closure criteria

- Customer sees one valid charge or two clearly differentiated legitimate transactions.
- Refund/reversal reference is recorded when applicable.
- Customer communication matches the actual transaction state.

Escalation

- Escalate large-value or disputed transactions beyond resolver authority.
- Escalate suspected fraud to Fraud Investigation.

OPERATOR NOTE
Never paste full payment card numbers or authentication data into the ticket or comments.

8. KB-FIN-002 - Subscription paid but service not activated

Use when payment succeeded but the purchased entitlement is not active.

Field	Value
Article ID	KB-FIN-002
Audience	Finance Operations, Technical Support
Severity / Priority	High
Applies when	Payment is confirmed but expected service remains unavailable.
Related articles	KB-FIN-001, KB-TECH-002

Resolution context

- Symptom: Payment is confirmed but expected service remains unavailable.
- Symptom: Customer has a valid order/subscription reference.
- Likely cause: Entitlement provisioning delay.
- Likely cause: Payment-to-subscription synchronization failure.
- Likely cause: Incorrect account mapping.
- Likely cause: Activation job failure.

Prerequisites and guardrails

- Confirm payment using approved transaction/reference data.
- Verify the target account and purchased plan.
- Do not manually alter entitlement records outside approved operations.

Resolution procedure

31. 1. Verify transaction status and subscription/order identifier.
32. 2. Confirm the intended customer account and plan.
33. 3. Check whether the entitlement record exists and whether activation completed.
34. 4. If a provisioning job failed, route to the owning technical queue.
35. 5. If payment is valid but entitlement is absent, coordinate finance and technical teams.
36. 6. Communicate the next expected milestone to the customer.

Validation and closure criteria

- Service entitlement is active.
- Payment and subscription records reconcile.
- Customer confirms access or the platform confirms successful activation.

Escalation

- Escalate provisioning/systemic activation failures.
- Escalate payment disputes through Finance Operations.

OPERATOR NOTE
Do not solve a finance reconciliation problem by changing technical entitlements without matching transaction evidence.

9. KB-FIN-003 - Refund request after service failure

Use when a customer requests a refund following service disruption or dissatisfaction.

Field	Value
Article ID	KB-FIN-003
Audience	Finance Operations, Customer Support
Severity / Priority	Medium to High
Applies when	Customer requests full or partial refund.
Related articles	KB-FIN-001, KB-FIN-002

Resolution context

- Symptom: Customer requests full or partial refund.
- Symptom: Service issue or failed delivery is documented.
- Likely cause: Service not delivered.
- Likely cause: Duplicate or erroneous charge.
- Likely cause: Customer goodwill request within policy.

Prerequisites and guardrails

- Follow contractual and refund-policy boundaries.
- Resolver may document evidence but should not approve refunds outside authority.

Resolution procedure

37. 1. Review ticket history and service-impact evidence.
38. 2. Identify transaction/reference and payment status.
39. 3. Determine eligibility under the applicable refund policy.
40. 4. Obtain required approval when outside resolver authority.
41. 5. Initiate approved refund workflow.
42. 6. Record refund amount, reference and communication.

Validation and closure criteria

- Refund request is approved/rejected with recorded rationale.
- Financial transaction reference is captured.
- Customer receives accurate outcome and timeline.

Escalation

- Escalate exceptions, high-value refunds or policy disputes.

OPERATOR NOTE
Separate service remediation from financial approval; both may be required on the same ticket.

10. KB-TECH-001 - Portal unavailable / broad service outage

Use when multiple users cannot access the application or a major service is unavailable.

Field	Value
Article ID	KB-TECH-001
Audience	Site Reliability, Technical Support, Incident Management
Severity / Priority	Critical to High
Applies when	Multiple users report the same outage.
Related articles	KB-TECH-002, KB-OPS-003

Resolution context

- Symptom: Multiple users report the same outage.
- Symptom: Core application or portal is unavailable.
- Likely cause: Infrastructure outage.
- Likely cause: Deployment/configuration regression.
- Likely cause: Dependency failure.
- Likely cause: Capacity saturation.

Prerequisites and guardrails

- Treat widespread impact as a potential incident, not just an individual ticket.
- Do not repeatedly retry destructive operations.
- Follow approved incident communication channels.

Resolution procedure

43. 1. Determine scope: one user, one tenant, multiple tenants or global.
44. 2. Compare ticket timing and symptoms across related cases.
45. 3. Check known incidents/platform health where available.
46. 4. Route to Site Reliability for broad availability failures.
47. 5. Capture customer business impact for incident prioritization.
48. 6. Keep the ticket linked to the parent incident when applicable.

Validation and closure criteria

- Service health is restored or incident owner confirms stabilization.
- Customer receives accurate status communication.
- Ticket references the parent incident where applicable.

Escalation

- Immediate incident escalation for broad/multi-tenant impact.
- Escalate critical business processes such as payroll or security functions.

OPERATOR NOTE
Avoid treating repeated customer reports as isolated tickets when evidence indicates a shared incident.

11. KB-TECH-002 - Business-critical integration/API failure

Use when an integration, API or automation required for business processing is failing.

Field	Value
Article ID	KB-TECH-002
Audience	Technical Support, Integration Team
Severity / Priority	High to Critical
Applies when	API requests fail consistently.
Related articles	KB-TECH-001, KB-OPS-001

Resolution context

- Symptom: API requests fail consistently.
- Symptom: Business process is blocked by an integration.
- Likely cause: Authentication/credential expiry.
- Likely cause: Schema or contract change.
- Likely cause: Endpoint availability failure.
- Likely cause: Rate limit or dependency issue.
- Likely cause: Application configuration mismatch.

Prerequisites and guardrails

- Use non-sensitive request/response identifiers.
- Do not expose production secrets in tickets.
- Check whether an approved maintenance or change is already in progress.

Resolution procedure

49. 1. Capture endpoint/service name and approximate failure window.
50. 2. Determine whether failures are all requests or a subset.
51. 3. Check response class or error signature if available.
52. 4. Validate configuration and credentials through approved controls.
53. 5. Compare the current interface contract against the supported version.
54. 6. Escalate to the integration/platform owner if the issue is systemic or persists after configuration validation.

Validation and closure criteria

- Successful request or business transaction completes.
- No credentials or sensitive payloads were exposed.
- Customer/business owner confirms recovery.

Escalation

- Escalate systemic integration defects, repeated 5xx responses or data integrity concerns.

OPERATOR NOTE
The resolver should distinguish an authentication/configuration issue from a service outage before routing to infrastructure.

12. KB-TECH-003 - Connectivity / intermittent access issue

Use when a user reports intermittent connectivity or access that does not appear to affect all users.

Field	Value
Article ID	KB-TECH-003
Audience	Technical Support
Severity / Priority	Medium to High
Applies when	Intermittent access.
Related articles	KB-TECH-001

Resolution context

- Symptom: Intermittent access.
- Symptom: Single user or limited group affected.
- Likely cause: Local network path.
- Likely cause: Client/browser cache.
- Likely cause: VPN/proxy behavior.
- Likely cause: Endpoint-specific configuration.

Prerequisites and guardrails

- Avoid requesting sensitive device credentials.
- Use customer-provided timestamps and safe diagnostic observations.

Resolution procedure

1. 1. Establish whether the issue follows the user, device or network.
2. 2. Compare browser/client behavior where appropriate.
3. 3. Clear safe client state only with customer approval.
4. 4. Test through an approved alternate network path where practical.
5. 5. Check whether known incidents exist.
6. 6. Escalate if multiple users share the same path or if service-side evidence appears.

Validation and closure criteria

- Stable connectivity is restored or evidence identifies a service-side issue.
- Resolution steps are documented.

Escalation

- Escalate shared network/service symptoms to Site Reliability.

OPERATOR NOTE
Avoid excessive local troubleshooting when multiple customers report the same issue.

13. KB-TECH-004 - Payroll or high-impact business process stopped

Use for complaints where a core business process such as payroll is stopped or materially delayed.

Field	Value
Article ID	KB-TECH-004
Audience	Technical Support, Site Reliability, Operations Manager
Severity / Priority	Critical
Applies when	Business-critical processing has stopped.
Related articles	KB-TECH-001, KB-TECH-002

Resolution context

- Symptom: Business-critical processing has stopped.
- Symptom: Large employee/customer population affected.
- Likely cause: Service outage.
- Likely cause: Integration failure.
- Likely cause: Data processing job failure.
- Likely cause: Dependency unavailable.

Prerequisites and guardrails

- Capture exact business impact and affected volume.
- Do not manually manipulate financial or payroll records without authorized procedure.

Resolution procedure

61. 1. Confirm business process, affected population and last successful run.
62. 2. Determine whether the issue is isolated to a tenant or systemic.
63. 3. Check parent incident/change status.
64. 4. Route to the technical owner based on the failure signature.
65. 5. Record recovery milestones and business-impact updates.
66. 6. Escalate to Operations Manager for sustained critical impact.

Validation and closure criteria

- Business process completes successfully or incident owner confirms recovery.
- Customer/business owner receives a clear recovery statement.
- Any manual remediation is documented and approved.

Escalation

- Immediate escalation for large employee/customer impact or prolonged outage.

OPERATOR NOTE
Business impact can justify Critical priority even when the technical symptom looks routine.

14. KB-OPS-001 - Ticket is unassigned / no automatic resolver

Use when AI classification succeeds but no eligible resolver is available.

Field	Value
Article ID	KB-OPS-001
Audience	Service Desk, Operations Admin
Severity / Priority	Operational
Applies when	Ticket is OPEN and unassigned.
Related articles	KB-OPS-002, KB-OPS-003

Resolution context

- Symptom: Ticket is OPEN and unassigned.
- Symptom: Assignment analysis reports no eligible resolver.
- Likely cause: Outside working shift.
- Likely cause: Resolver at capacity.
- Likely cause: No qualified skill match.
- Likely cause: Team taxonomy mismatch.
- Likely cause: All normal workforce candidates unavailable.

Prerequisites and guardrails

- Do not bypass the operational team boundary.
- Use manual assignment only to an active resolver in the correct operational team unless an approved exception exists.

Resolution procedure

67. 1. Open assignment analysis and review all candidate explanations.
68. 2. Confirm the operational team resolved from the ticket classification.
69. 3. Review workforce status, shift, capacity and skill signals.
70. 4. If an eligible resolver exists, use auto-assignment.
71. 5. If no automatic candidate exists, use authorized manual assignment.
72. 6. Record the reason for the override.
73. 7. If no resolver exists in the team, escalate to workforce/queue administration.

Validation and closure criteria

- Ticket has an owner.
- Assignment method and reason are captured.
- Ticket remains OPEN until the resolver actually starts work.

Escalation

- Escalate if a required operational team has no active resolver.
- Escalate critical queue starvation immediately.

OPERATOR NOTE
Automatic routing failure is not a platform failure by itself; it may be the expected result of workforce constraints.

15. KB-OPS-002 - SLA breached / SLA at risk

Use when a ticket is approaching or has exceeded its response or resolution SLA.

Field	Value
Article ID	KB-OPS-002
Audience	Service Desk, Resolver, Operations Manager
Severity / Priority	High to Critical
Applies when	SLA state is AT_RISK or BREACHED.
Related articles	KB-OPS-001, KB-OPS-004

Resolution context

- Symptom: SLA state is AT_RISK or BREACHED.
- Symptom: Customer impact may be increasing because of delay.
- Likely cause: No resolver ownership.
- Likely cause: High workload on resolver.
- Likely cause: Customer dependency.
- Likely cause: Queue starvation.
- Likely cause: Underlying incident.

Prerequisites and guardrails

- Use the platform SLA state as the source of truth.
- Do not reset or manually manipulate SLA deadlines without authorized policy change.

Resolution procedure

74. 1. Inspect first-response and resolution SLA separately.
75. 2. Determine whether the ticket is assigned and actively worked.
76. 3. Identify blocker: resolver capacity, customer response, technical dependency or queue condition.
77. 4. Escalate according to priority and SLA policy.
78. 5. Notify customer when an approved service communication is required.
79. 6. Record the mitigation or escalation.

Validation and closure criteria

- SLA risk has been reduced or ownership is transferred appropriately.
- Customer communication is accurate.
- Escalation and audit records are complete.

Escalation

- Critical breached tickets require immediate operational attention.
- Escalate repeated queue-level breaches as a workforce/process problem.

OPERATOR NOTE
SLA breach does not automatically mean the ticket is technically unresolved; it means the defined service commitment was missed.

16. KB-OPS-003 - Ticket wrongly assigned or needs reassignment

Use when the ticket is assigned to the wrong resolver/team or requires operational rebalancing.

Field	Value
Article ID	KB-OPS-003
Audience	Operations Admin, Operations Manager
Severity / Priority	Operational
Applies when	Assigned resolver lacks required skills.
Related articles	KB-OPS-001, KB-AI-001

Resolution context

- Symptom: Assigned resolver lacks required skills.
- Symptom: Team assignment is incorrect.
- Symptom: Resolver is overloaded or unavailable.
- Likely cause: Classification drift.
- Likely cause: Legacy team naming.
- Likely cause: Workforce state changed after assignment.
- Likely cause: Manual assignment error.

Prerequisites and guardrails

- Keep the original assignment event in the audit trail.
- Do not erase assignment history.
- Use reassignment with a documented operational reason.

Resolution procedure

80. 1. Open assignment analysis for the ticket.
81. 2. Compare current resolver skills and workload with the ticket requirement.
82. 3. Verify the canonical operational team.
83. 4. Identify the strongest eligible replacement.
84. 5. Reassign through the approved workflow.
85. 6. Confirm the new resolver sees the ticket in their queue.
86. 7. Notify affected parties when notification policy requires it.

Validation and closure criteria

- New resolver owns the ticket.
- Audit shows previous and new resolver.
- No duplicate active work remains.

Escalation

- Escalate repeated misrouting to routing-model/taxonomy ownership.

OPERATOR NOTE
Do not correct a routing error by weakening team or skill constraints. Fix the routing signal or use a documented manual override.

17. KB-OPS-004 - Escalated ticket requiring operations-manager review

Use when a ticket has reached an escalation condition or repeated SLA breach and requires management attention.

Field	Value
Article ID	KB-OPS-004
Audience	Operations Manager, Service Desk
Severity / Priority	High to Critical
Applies when	Escalation level increased.
Related articles	KB-OPS-002, KB-TECH-001, KB-SEC-002

Resolution context

- Symptom: Escalation level increased.
- Symptom: Ticket has repeated SLA breaches or major customer impact.
- Likely cause: SLA breach.
- Likely cause: Critical priority.
- Likely cause: High business impact.
- Likely cause: Resolver or team capacity problem.

Prerequisites and guardrails

- Review the complete ticket timeline before making an override decision.
- Maintain customer communication discipline.

Resolution procedure

87. 1. Review ticket summary, complaint, priority, SLA, assignment and audit history.
88. 2. Confirm whether the escalation owner is correct.
89. 3. Decide whether to reassign, increase capacity, engage another team or declare/associate an incident.
90. 4. Record management decision and owner.
91. 5. Ensure the resolver and customer receive required notifications.
92. 6. Continue monitoring until the risk is closed.

Validation and closure criteria

- Escalation owner is identified.
- Decision and actions are documented.
- SLA and customer-impact risk are actively managed.

Escalation

- Escalate to incident/security/privacy governance when the issue exceeds service-desk authority.

OPERATOR NOTE
Escalation should change the operational response, not merely change a numeric level.

18. KB-AI-001 - Using the AI resolution recommendation safely

Use when a resolver requests AI-generated investigation and resolution guidance.

Field	Value
Article ID	KB-AI-001
Audience	Resolver, Operations Manager
Severity / Priority	Assistive / advisory
Applies when	Ticket is active and requires troubleshooting guidance.
Related articles	KB-AI-002, KB-SEC-001

Resolution context

- Symptom: Ticket is active and requires troubleshooting guidance.
- Symptom: Resolver wants a recommended investigation path.
- Likely cause: Complex or unfamiliar issue.
- Likely cause: Need to synthesize ticket context and prior cases quickly.

Prerequisites and guardrails

- AI output is advisory, not proof.
- Never treat AI output as evidence that a system action succeeded.
- Use approved system data, KB guidance and historical cases to validate recommendations.

Resolution procedure

93. 1. Review ticket context before generating the recommendation.
94. 2. Generate the AI recommendation.
95. 3. Compare the recommendation with approved KB/SOP procedures.
96. 4. Check similar resolved cases for precedent and differences.
97. 5. Execute only approved actions.
98. 6. Record material remediation in the ticket.
99. 7. Re-run or refine the recommendation only when new evidence changes the diagnosis.

Validation and closure criteria

- Resolver validates the recommendation against real system evidence.
- Resolution steps are documented.
- Customer response does not contain unverified technical claims.

Escalation

- Escalate when AI advice conflicts with policy, evidence or security controls.
- Escalate high-risk actions rather than treating AI confidence as authorization.

OPERATOR NOTE
AI should narrow investigation effort, not replace operational accountability.

19. KB-AI-002 - Using similar resolved tickets as precedent

Use when the resolver wants historical examples before deciding how to resolve a current complaint.

Field	Value
Article ID	KB-AI-002
Audience	Resolver, Service Desk
Severity / Priority	Assistive / advisory
Applies when	The platform returns one or more similar RESOLVED/CLOSED cases.
Related articles	KB-AI-001, KB-OPS-003

Resolution context

- Symptom: The platform returns one or more similar RESOLVED/CLOSED cases.
- Likely cause: Repeated complaint pattern.
- Likely cause: Known operational symptom.
- Likely cause: Need for prior successful remediation pattern.

Prerequisites and guardrails

- Similarity is evidence for consideration, not a guarantee that two incidents have the same root cause.
- Compare customer context, environment, version, timing and business impact.

Resolution procedure

100. 1. Review the top similar cases.
101. 2. Compare symptom and category alignment.
102. 3. Identify the resolution pattern that was successful.
103. 4. Check whether the original conditions are still valid.
104. 5. Reuse safe troubleshooting steps when applicable.
105. 6. Document deviations when the current ticket differs from precedent.

Validation and closure criteria

- Chosen remediation is validated against current evidence.
- Historical precedent is referenced where useful.

Escalation

- Escalate if historical precedent conflicts with current platform behavior or security policy.

OPERATOR NOTE
Prefer approved historical patterns over generic internet advice, but do not blindly replay old actions.

20. Operational Playbook - Assignment Troubleshooting Matrix

Observed condition	Automatic routing outcome	Admin action	What not to do
No eligible resolver	Ticket remains OPEN/unassigned	Review assignment analysis; assign manually if policy allows	Do not weaken team boundary globally
Correct team, wrong skills	Candidate filtered or scored low	Select a qualified resolver or correct taxonomy	Do not assign unrelated specialist just to clear queue
Resolver at capacity	Automatic candidate rejected or penalized	Rebalance workload or use documented override	Do not silently exceed capacity
Outside working shift	Normal routing rejects candidate	Use eligible resolver or approved on-call path	Do not assume AVAILABLE means currently on shift
Critical + no normal resolver	On-call fallback may engage	Review emergency coverage and escalation	Do not treat on-call as normal queue
Legacy team label	Canonical team mapping resolves it	Confirm operational team before manual assignment	Do not create duplicate teams for spelling variants

20.1 Workforce interpretation

- AVAILABLE is a status; it does not necessarily mean the resolver is in an active shift.
- eligible_for_assignment is the final workforce decision for automatic routing.
- capacity_available and utilization must be interpreted together with active-ticket count.
- effective workload may be higher than raw ticket count because priority weighting is part of routing intelligence.

OPERATIONAL PATTERN
When automatic routing skips a ticket, the administrator should be told the specific constraint, not only the phrase “No eligible resolver.” The message should identify team, workforce and capacity causes so manual action is explainable.

21. Operational Playbook - SLA and Escalation

Situation	Immediate action	Escalation path	Evidence to record
Critical ticket created	Confirm team and resolver; monitor first-response SLA	Operations Manager / incident owner when required	Assignment reason, SLA state
SLA at risk	Identify blocker; increase attention or rebalance	Team lead / operations manager	Blocker, mitigation, owner
Resolution SLA breached	Confirm active ownership and customer communication	Operations Manager when material	Breach state, cause, action
Repeated breaches in one queue	Treat as systemic workforce/process issue	Workforce + operations leadership	Queue trend, capacity, staffing
Security/privacy impact	Contain and preserve evidence	Cyber Security / Privacy	Incident reference, evidence scope
Customer dependency	Move to PENDING_CUSTOMER when appropriate	Follow up per communication policy	Question asked and required customer action

21.1 SLA handling rules

- Use the platform SLA state rather than local wall-clock calculations.
- Separate first-response SLA from resolution SLA; one may breach while the other remains healthy.
- SLA pause state must be visible in the audit trail where the process permits pausing.
- A resolved ticket should stop active resolution timing according to the platform lifecycle rules.

SLA PRINCIPLE
The SLA engine is a control mechanism, not merely a countdown display. It should drive prioritization, assignment attention, escalation and operational reporting.

22. Resolver Checklist - Before You Resolve

106. Confirm the requested outcome is understood and matches the complaint.
107. Review AI recommendation and similar resolved cases only as supporting context.
108. Confirm the actual technical/business state using approved evidence.
109. Verify customer impact is addressed.
110. Record the material resolution steps taken.
111. Confirm no related security, privacy, financial or incident-management obligation remains open.
112. Move to RESOLVED only when the resolver has completed the work and the evidence supports the outcome.
113. Provide a concise customer-facing resolution statement.

22.1 Quality checklist

Check	Pass condition
Identity / customer context	Customer identity and account context are correctly associated
Root cause	Root cause or most supported explanation is documented
Action	Actual action performed is recorded
Validation	Successful outcome is verified
SLA	SLA state and any breach reason are understood
Security / privacy	No unresolved security/privacy concern remains
Communication	Customer has a clear outcome and next step if any
Audit	Material assignment, status, escalation and comment events exist

RESOLVE VS. CLOSE
RESOLVED means the resolver has completed the service work. CLOSED is the administrative end state. Do not use closure to hide an unresolved technical condition.

23. Knowledge Base Governance and Article Quality

23.1 Article lifecycle

Stage	Owner	Output
Draft	Resolver subject-matter owner	Article created with evidence and safe procedure
Review	Operations + technical SME	Technical accuracy, scope and escalation validated
Approve	Knowledge owner / delegated authority	Article becomes controlled
Publish	Platform administrator	Searchable and versioned article available
Measure	Knowledge owner	Usage, helpfulness and resolution outcomes reviewed
Retire	Knowledge owner	Superseded article removed from active retrieval

23.2 Article quality gates

- Every article must have a stable ID and owner.
- Steps must be actionable and sequenced.
- Every article must state when to stop and escalate.
- Sensitive data handling rules must be explicit for security, identity and finance articles.
- Validation criteria must be measurable or observable.
- Related articles should connect common incident chains.
- Major product/process changes should trigger article review.

23.3 AI retrieval readiness

For future retrieval-augmented generation or GraphRAG use, preserve entities and relationships explicitly: product area, category, subcategory, team, resolver role, symptom, cause, procedure, validation, escalation owner and related articles. This allows the platform to retrieve a precise operational path rather than a generic troubleshooting paragraph.

DESIGN RULE
The KB should be the approved procedural source. Historical tickets are evidence of what happened; KB articles are the controlled source of what should be done.

24. Search and Retrieval Design Reference

24.1 Recommended query sequence

114. Retrieve by ticket category and subcategory.
115. Filter by resolver role and applicable system domain.
116. Rank by symptom/complaint similarity.
117. Prefer current-version articles.
118. Retrieve similar RESOLVED/CLOSED tickets as evidence.
119. Present AI guidance with source references in the resolver experience.

24.2 Suggested metadata dictionary

Field	Purpose	Example
article_id	Stable knowledge identifier	KB-SEC-001
domain	High-level operational domain	Security & Privacy
category	Ticket/incident category	Security
subcategory	Specific issue pattern	Account Compromise
symptom	Observable issue	Suspicious login alerts
priority	Expected urgency	Critical
audience	Allowed resolver roles	Cyber Security
keywords	Retrieval terms	credential, suspicious login
procedure	Approved resolution steps	Ordered procedural text
validation	Completion criteria	Sessions contained; reset completed
escalation	Stop conditions and owner	Cyber Security incident process
related_articles	Graph/relationship links	KB-FIN-001, KB-SEC-002

24.3 Why this structure matters

- It supports lexical and semantic retrieval.
- It reduces hallucination risk by constraining the AI to controlled procedures.
- It enables graph relationships between incidents, teams, SLAs and remediation patterns.
- It makes KB analytics possible: which articles resolve tickets, which articles are stale and which topics generate repeated escalations.

25. Appendix A - Common Diagnostic Questions

Issue family	Questions to ask first
Account access	What exact error? When did it start? One user or many? Any recent password/security change?
Security	What alert occurred? When? Is the account still accessible? Any unauthorized transaction or data exposure?
Billing	What transaction/reference? Amount/date/currency? Pending or settled? Duplicate or unexpected?
Subscription	Was payment confirmed? What plan? What account? Is entitlement absent or inactive?
Outage	How many users/tenants affected? What was the last successful operation? Any known incident?
Integration/API	Endpoint/service? Failure time? Error class/signature? All requests or subset? Recent change?
SLA	Which SLA? First response or resolution? What is blocking progress? Is a resolver assigned?
Routing	What team did AI select? Are there eligible resolvers? Is the ticket legacy-labeled? Is manual assignment required?

DO NOT ASK FOR
Passwords, MFA codes, private recovery secrets, full payment card numbers or unrelated personal data. Use approved verification and evidence-handling procedures.

26. Appendix B - Quick Reference: Enterprise Resolution Flow

Use this one-page flow when a resolver needs a fast decision path without reading a full article.

Step	Decision	Action
1. Intake	What is the user actually experiencing?	Read complaint, category, subcategory and business impact.
2. Priority	How urgent is the impact?	Use platform priority; verify customer-requested priority does not override evidence-based risk.
3. Team	Who should own this domain?	Use canonical operational team.
4. Resolver	Who should work it?	Use workforce-aware assignment; review skills, workload, capacity and SLA health.
5. Diagnose	What is the most likely cause?	Use approved KB + similar resolved tickets + AI recommendation.
6. Act	What approved step changes the condition?	Execute only documented and authorized actions.
7. Validate	Did the condition actually change?	Confirm through system evidence or customer confirmation.
8. Communicate	What does the customer need to know?	State outcome, limitations and next steps.
9. Resolve	Is service work complete?	Move to RESOLVED only after evidence supports closure.
10. Close	Has administrative closure criteria been met?	Close after required confirmations and audit records.

FINAL CONTROL
When the correct action is uncertain, stop at diagnosis and escalate. A controlled delay is safer than an unverified remediation.

27. Appendix C - Article Index

Article ID	Title	Severity	Audience
KB-ACC-001	Account locked / unable to sign in	['Customer reports locked account or repeated login rejection.', 'No evidence yet of malicious account takeover.']	['Account lockout policy triggered.', 'Incorrect credentials or stale browser/session state.', 'Identity synchronization or authentication service issue.']
KB-ACC-002	Password reset email not received	['Customer can access email but no reset message is received.', 'Reset request may succeed in the application but email is absent.']	['Mailbox filtering or spam quarantine.', 'Email delivery delay.', 'Incorrect customer email or stale identity record.', 'Notification provider outage.']
KB-SEC-001	Suspicious login alert / suspected credential compromise	['Unexpected login from an unfamiliar location.', 'Customer believes credentials were compromised.']	['Credential theft.', 'Session/token compromise.', 'Compromised endpoint or browser.']
KB-SEC-002	Customer sees another customer's personal information	['Customer can view another user/customer record.', 'Screenshot or evidence indicates unauthorized data access.']	['Authorization policy defect.', 'Tenant/row-level data isolation failure.', 'Cache or session contamination.', 'Configuration or deployment regression.']
KB-FIN-001	Duplicate charge / duplicate payment	['Two or more transactions appear to represent the same intended purchase.', 'Customer requests reversal or refund.']	['Duplicate submission.', 'Payment retry after timeout.', 'Gateway duplication.', 'Delayed authorization/capture creating apparent duplicates.']
KB-FIN-002	Subscription paid but service not activated	['Payment is confirmed but expected service remains unavailable.', 'Customer has a valid order/subscription reference.']	['Entitlement provisioning delay.', 'Payment-to-subscription synchronization failure.', 'Incorrect account mapping.', 'Activation job failure.']
KB-FIN-003	Refund request after service failure	['Customer requests full or partial refund.', 'Service issue or failed delivery is documented.']	['Service not delivered.', 'Duplicate or erroneous charge.', 'Customer goodwill request within policy.']
KB-TECH-001	Portal unavailable / broad service outage	['Multiple users report the same outage.', 'Core application or portal is unavailable.']	['Infrastructure outage.', 'Deployment/configuration regression.', 'Dependency failure.', 'Capacity saturation.']
KB-TECH-002	Business-critical integration/API failure	['API requests fail consistently.', 'Business process is blocked by an integration.']	['Authentication/credential expiry.', 'Schema or contract change.', 'Endpoint availability failure.', 'Rate limit or dependency issue.', 'Application configuration mismatch.']
KB-TECH-003	Connectivity / intermittent access	['Intermittent access.', 'Single user	['Local network path.',

Article ID	Title	Severity	Audience
	issue	or limited group affected.']	'Client/browser cache.', 'VPN/proxy behavior.', 'Endpoint-specific configuration.']
KB-TECH-004	Payroll or high-impact business process stopped	['Business-critical processing has stopped.', 'Large employee/customer population affected.']	['Service outage.', 'Integration failure.', 'Data processing job failure.', 'Dependency unavailable.']
KB-OPS-001	Ticket is unassigned / no automatic resolver	['Ticket is OPEN and unassigned.', 'Assignment analysis reports no eligible resolver.']	['Outside working shift.', 'Resolver at capacity.', 'No qualified skill match.', 'Team taxonomy mismatch.', 'All normal workforce candidates unavailable.']
KB-OPS-002	SLA breached / SLA at risk	['SLA state is AT_RISK or BREACHED.', 'Customer impact may be increasing because of delay.']	['No resolver ownership.', 'High workload on resolver.', 'Customer dependency.', 'Queue starvation.', 'Underlying incident.']
KB-OPS-003	Ticket wrongly assigned or needs reassignment	['Assigned resolver lacks required skills.', 'Team assignment is incorrect.', 'Resolver is overloaded or unavailable.']	['Classification drift.', 'Legacy team naming.', 'Workforce state changed after assignment.', 'Manual assignment error.']
KB-OPS-004	Escalated ticket requiring operations-manager review	['Escalation level increased.', 'Ticket has repeated SLA breaches or major customer impact.']	['SLA breach.', 'Critical priority.', 'High business impact.', 'Resolver or team capacity problem.']
KB-AI-001	Using the AI resolution recommendation safely	['Ticket is active and requires troubleshooting guidance.', 'Resolver wants a recommended investigation path.']	['Complex or unfamiliar issue.', 'Need to synthesize ticket context and prior cases quickly.']
KB-AI-002	Using similar resolved tickets as precedent	['The platform returns one or more similar RESOLVED/CLOSED cases.']	['Repeated complaint pattern.', 'Known operational symptom.', 'Need for prior successful remediation pattern.']

This index is intentionally suitable for later migration into a knowledge-base table or API. Each article can become a record with content chunks plus metadata while retaining the stable Article ID.

IMPLEMENTATION NOTE
For the complaint-routing platform, the next logical ingestion target is a structured KB service with article CRUD, semantic search, versioning, and retrieval hooks for the resolver AI assistant. The controlled DOCX/PDF remains the human-readable master document.